

BRIDGE 5722

Carries North Section Street (U.S. Highway 63/Trunk Highway 1)
over Spring Valley Creek
Spring Valley
Fillmore County
Minnesota

HAER MN-125

HAER MN-125

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

HISTORIC AMERICAN ENGINEERING RECORD

National Park Service
U.S. Department of the Interior
1849 C Street NW
Washington, DC 20240-0001

HISTORIC AMERICAN ENGINEERING RECORD

BRIDGE NO. 5722

HAER No. MN-125

Location: Carries North Section Street (U.S. Highway 63/Trunk Highway 1) over Spring Valley Creek, city of Spring Valley, Fillmore County, Minnesota

Bridge No. 5722 is located at latitude 43.688041, longitude -92.389099. The coordinate represents the center of the structure. It was obtained in 2014 by plotting its location in Google Earth. The location has no restriction on its release to the public.

Present Owner/ Occupant: Minnesota Department of Transportation.

Present Use: Vehicular and pedestrian traffic.

Significance: Bridge No. 5722 is an unusual engineering adaptation of the Minnesota Department of Highway's (MHD) standardized box culvert plan for use on a U.S. Highway to meet site-specific conditions normally accommodated by a bridge design. Built in 1936, Bridge No. 5722 is only one of three extant examples on U.S. Highways in Minnesota of a concrete box culvert design adapted by the MHD to function as a bridge and meet the requirements of both vehicular and pedestrian traffic.

Historian(s): Katherine Haun and Amy Squitieri; Mead & Hunt, Inc.; 2014.

Project Information: This documentation was prepared as mitigation for the Advisory Council on Historic Preservation (ACHP)'s *Program Comment Issued for Streamlining Section 106 Review for Actions Affecting Post-1945 Concrete and Steel Bridges*, noticed in the Federal Register on November 16, 2012. The Federal Highway Administration (FHWA) and the ACHP are the project sponsors and retained the team of Louis Berger and Mead & Hunt, Inc. to prepare this document. It was prepared by Katherine Haun, Cultural Resources Specialist, Mead & Hunt, Inc. Dietrich Floeter completed the photography.

Part I. Historical Information

A. Physical History:

1. Date(s) of construction: 1936.

2. Engineer: Minnesota Department of Highways.

3. Builder/Contractor/Supplier: J.S. McLaughlin & Sons, Mankato, Minnesota, contractors; S.O. Koland & Son, [unknown location], subcontractor.

4. Original plans and construction: The original plans, "Special Triple 12 Ft. x 6 Ft Box Culvert," prepared by the Minnesota Highway Department, are dated June 26, 1936.¹ Construction was completed the same year.²

5. Alterations and additions: Repair work, including repairs to the culvert cells, sidewalks, and deck, has been minor. In 1996 new concrete endposts that replicate the original design were installed, sidewalks and curbs were reconstructed, and the culvert ceilings were repaired. MnDOT repaired cracks in the roadway in 2012.³ At an unknown time, the southwest wingwall was altered with additional concrete on the diagonal top edge of the wingwall to bring the wingwall to sidewalk height and allow for a power pole to be added. This wingwall now abuts a retaining wall for the south bank of Spring Valley Creek. There have been no other alterations to the culvert or wingwalls.

B. Historical Context:

Development of Trunk Highway 59/U.S. Highway 63 in Spring Valley

Bridge No. 5722 is located on U.S. Highway (US) 63 (formerly Trunk Highway [TH] 59) in the community of Spring Valley in Fillmore County, Minnesota.

¹ While Bridge No. 5722 was designed using the 1932 standard plan, because the bridge was skewed 35 degrees it is known as a "Special Triple 12 Ft. x 6 Ft. Box Culvert." See sheet 1, "Special Triple 12 Ft x 6 Ft Box Culvert," State of Minnesota Department of Highways, "Bridge 5722 Plans," 26 June 1936, Minnesota Department of Transportation electronic documents, http://dotapp7.dot.state.mn.us/cyberdocs_guest/Libraries/Default_Library/Groups/GUESTS/frameset.asp (accessed 17 February 2014).

² E.J. Miller, photograph of completed Bridge No. 5722, 5 November 1936, in Bridge No. 5722 envelope, Box 4, Bridge Photographs and Negatives, Minnesota Highway Department Bridge Division, Minnesota State Archives, Minnesota Historical Society, St. Paul, Minn.

³ Bridge No. 5722 Correspondence File, available at the Bridge Office, Minnesota Department of Transportation, St. Paul, Minn.

The Minnesota Highway Commission (later known as the Minnesota Department of Highways or MHD) implemented the development of the Trunk Highway system in 1921, a state system of improved roads that provided efficient automobile transportation between communities.⁴ Due to the increased need for rural mail delivery and the growing use of roads in the winter, the MHD announced in November 1926 that winter traffic was “sufficient to fully justify us spending the money” on snow plows, graders, and highway improvements in Fillmore County.⁵ By 1928 two major regional roads ran through Spring Valley: the road that would become TH 59 and eventually US 63, which followed a north-south route through the center of Spring Valley, and TH 16, which curved through the village at a northwest to southeast angle.⁶

Ownership and maintenance of the roadway and bridges within Spring Valley were assumed by the MHD in 1931. North Section Street was added to the state Trunk Highway system as part of TH 59, with the bridge over Spring Valley Creek receiving the state bridge number of M-1125.⁷ An extension of the existing US 63 from Iowa to Rochester, Minnesota, was proposed by the MHD in July 1933. In 1928, only a few years earlier, the Mayo Clinic had completed construction of the Plummer Building, now a city landmark, symbolizing Rochester’s rise to national, and even international, prominence as a medical destination.⁸ The proposed routing would upgrade TH 59 from the Iowa-Minnesota border north through Spring Valley to Rochester.⁹ TH 59 became US 63 in 1935.

With the upgrade of TH 59 to a U.S. Highway, improvements to the road grade and associated bridges were necessary to provide greater width and capacity.¹⁰ As was the case of M-1125, many bridges that the MHD took over did not meet the highway standards and specifications and were upgraded or replaced.¹¹ Though the state owned the road and bridges through Spring Valley on TH 59, improvement plans were brought forward by E.W Krueger, MHD Engineer, to the Spring Valley Village Council for

⁴ National Register of Historic Places, Reinforced-Concrete Highway Bridges in Minnesota Multiple Property Documentation Form, State of Minnesota, National Register #64500291, E-12, G-1, available at Minnesota State Historic Preservation Office, St. Paul, Minn.

⁵ Spring Valley Historical Society, *Tales of Our Town – Spring Valley Sesquicentennial Stories 1855-2005* (Spring Valley, Minn.: n.p., 2005), 39.

⁶ Webb Publishing Co, *Atlas and Farmers’ Directory of Fillmore County, Minnesota: Containing Plats of All Townships with Owners’ Names* (St. Paul, Minn.: Webb Publishing Co., 1928).

⁷ Minnesota Department of Transportation, “Fillmore County, US/TH 63 Construction Project Log, Section 2313,” Minnesota Department of Transportation roadway data, <http://www.dot.state.mn.us/roadway/data/docs/projlog/county23/2313.pdf> (accessed 16 April 2014).

⁸ David Gebhard and Tom Martinson, *A Guide to the Architecture of Minnesota* (Minneapolis: University of Minnesota Press, 1977), 312, 314.

⁹ Spring Valley Historical Society, *Tales of Our Town*, 40.

¹⁰ Entry for 7 May 1936, Spring Valley City Council Minutes, Book 1934-1936, Spring Valley Records 1928-1980, Fillmore County, Minnesota State Archives, Minnesota Historical Society, St. Paul, Minn.

¹¹ National Register of Historic Places, Reinforced-Concrete Highway Bridges in Minnesota Multiple Property Documentation Form, E-13.

comment and approval on May 7, 1936.¹² The improvements involved widening an important roadway that connected the railroad depot to the commercial center of town. Village Council approval indicated the community was in agreement with proposed plans.

MHD's standardization of concrete box culverts and design of Bridge No. 5722

US 63 improvement plans included the replacement of M-1125, the original 1916 two-span bridge at this location, with a concrete box culvert.¹³ The Minnesota Highway Commission first released standardized plans for the construction of concrete box culverts in 1912.¹⁴ By 1936 Minnesota had 375 concrete box culverts out of approximately 8,900 bridges and structures inventoried by the MHD.¹⁵ Contemporary bridge engineering books, pamphlets, and technical circulars promoted the use of concrete culverts and other small structures as having the advantages of economy, stiffness, resistance to temperature and shrinkage, and ease of construction.¹⁶

The concrete box culvert chosen for this U.S. Highway crossing was not as conventional a choice as other bridge designs, such as a reinforced-concrete slab or beam bridge. While a conventional bridge design would seem to be appropriate to accomplish essentially the same task, a concrete box culvert turns out to have been a more efficient and logical choice for spanning the Spring Valley Creek at this particular point on TH 59. A concrete box culvert in its simplest form is a drainage opening beneath a roadway that is monolithic in construction, meaning that, unlike a bridge, there is no distinction between the substructure and superstructure. One economic benefit of the monolithic construction is that the bottom slab (the culvert's floor) serves as the structure's footing or foundation and no pilings are required as in a bridge to stabilize the structure in the subsoil. Another, and perhaps more important, benefit is that the top of culvert serves as the driving surface, without the need for a separate deck structure. In this way, the box culvert has a low profile that enables the road to be constructed without significantly raising its grade, as would have been required with a conventional bridge. In other words, the concrete box culvert was a simple, strong structure that was shallow enough to fit in the space between the existing

¹² Entry for 7 May 1936, Spring Valley City Council Minutes.

¹³ Entry for 10 July 1916, Board of County Commissioners, Fillmore County, Minutes, Book E, 156, Fillmore County Board of Commissioners Minutes and Agenda, Minnesota State Archives, Minnesota Historical Society, St. Paul, Minn.; M-1125 visible in E.J. Miller, photograph of M-1125, 17 April 1935, in Bridge No. 5722 envelope, Box 4, Bridge Photographs and Negatives, Minnesota Highway Department Bridge Division, Minnesota State Archives, Minnesota Historical Society, St. Paul, Minn.; Bridge No. 5722 is included in the National Inventory of Bridges (NIB) as a bridge, while structurally it is a concrete box culvert.

¹⁴ "Standard Triple 12 Ft x 6 Ft Box Culvert," approved by the MHD in 1932, was used as the basis for designing Bridge No. 5722. See "Standard Triple 12 Ft x 6 Ft Box Culvert," single plan sheet, 15 June 1932, Minnesota Highway Department, copy provided by MnDOT, Bridge Office, Oakdale, Minn.; National Register of Historic Places, Reinforced-Concrete Highway Bridges in Minnesota Multiple Property Documentation Form, E-12.

¹⁵ The MHD used Public Road Administration funds to administer the inventory of bridges within the state in 1935-36. See E.J. Miller, "Standard Bridge and Culvert Plans," *Better Roads* (June 1947): 23.

¹⁶ Parsons Brinkerhoff and Engineering and Industrial Heritage, "Context for Common Historic Bridge Types," prepared for the National Cooperative Highway Research Program, October 2005, 3-84.

grade of the highway and the creek bed, and used the simplest of foundations, essentially a spread footing. In the case of Bridge No. 5722, this also made for a more economical structure to build.

As the MHD engineer's survey notes for the planned culvert stated, Spring Valley Creek was known to flood, indicating flood-related issues for other nearby bridges in the 1930s. A bridge with adequate hydrological capacity was necessary, according to the survey plans, because "all bridges upstream are too small as high water leaves banks and floods the approach grades."¹⁷ To meet the hydraulic requirements of the Spring Valley Creek site, Bridge No. 5722 was designed as a three-cell culvert, notably larger than conventional culverts with one or two cells.¹⁸ Flooding of the streets was common during high water situations regardless of a higher profile road and bridge. While the box culvert might trap debris around the cell walls (corresponding to piers in a conventional bridge), it would be durable in high water situations and not as susceptible to "scour," or streambed erosion. A conventional bridge would have a more complex substructure with abutments and possibly a pier, any of which might be subject to scour and undermining. The choice of the three-cell box culvert allowed for a durable structure that could accommodate as much water as necessary without having to raise the road and construct an expensive, large bridge that would likely flood anyway.

Bridge No. 5722 was designed as a cast-in-place culvert. After World War II cast-in-place structures were largely superseded by pre-cast concrete units. Cast-in-place concrete culverts were constructed on-site; in comparison, pre-cast concrete culverts were manufactured off-site by a fabricator and brought to a particular location. Pre-cast concrete box culverts were not widely available until the 1970s, at which time standard designs and plans were developed by the American Society for Testing and Materials (ASTM) and the American Association of State Highway and Transportation Officials (AASHTO). The ability to use precast box culverts as both a conduit and a bridge-type structure reduced construction time, which in turn led to increased use by state departments of transportation.¹⁹

Concrete box culverts, like many small and simple structures built repeatedly throughout the state, used MHD-designed standard plans for efficiency. If necessary, the standard plans could be economically adapted to special circumstances with a minimum of new engineering. Bridge No. 5722 was based on a standard plan titled "Standard Triple 12 Ft. x 6 FT. Box Culvert," adopted by the MHD in June 1932, for a straightforward culvert with three cells, each 12'-0" wide and 6'-0" high. When adapted for the Spring

¹⁷ See Sheet 3, "Report of Bridge Survey," State of Minnesota Department of Highways, "Bridge 5722 Plans," 26 June 1936.

¹⁸ The typical box culvert has spans of three to 12 feet. U.S. Department of Transportation, *Bridge Inspectors Training Manual/90*, 19-4.

¹⁹ Arlen Ottman, email to Mead & Hunt, Inc. 21 February 2008; Timothy J. McGrath, et al. "Culverts: Hidden Structures, Critical Infrastructure Elements," Transportation Research Circular E-C104, (September 2006): 114-115.

Valley Creek crossing of US 63, the plan became "Special Triple 12 Ft. x 6 Ft Box Culvert" (also indicated as "Special W126T," culvert-shorthand for 12'-0" wide and 6'-0" tall) that was approved June 26, 1936.²⁰

This "special" adaptation effectively transformed a standard-plan box culvert into a highway bridge²¹ serving the needs of a newly upgraded state highway passing through a flood-prone section of a community with requirements for both vehicular and pedestrian traffic. The MHD avoided the engineering time and expense of new design that a conventional bridge would have required. As "Special W126T," Bridge No. 5722 included several accommodations to the site. It was skewed 35 degrees to accommodate the Spring Valley Creek crossing angle, it used the top of the culvert as the roadway surface, and it provided sidewalks and ornamental railings. The addition of a railing and sidewalks is unusual for a culvert. Typically, culverts are simple structures, with several feet or more of fill between the culvert and the road surface, without ornamentation, and without the need of a railing. The ornamental steel "Wisconsin Rail," as specified on the original survey sheet, and the 8'-0" sidewalks were added to meet the community's more "urban" pedestrian needs, since the structure connected Spring Valley's commercial district west of the highway with the railroad depot to the east.

The \$172,762 contract for TH 59 improvements through Fillmore County and within Spring Valley, including the culvert, was let by the MHD to J.S. McLaughlin & Sons of Mankato, Minnesota, on June 4, 1936.²² The work was funded under Federal Aid Project No. 409-C.²³ The culvert construction was subcontracted to S.O. Koland & Son.²⁴ Little attention was paid to the construction of Bridge No. 5722 by local news sources as it was simply a routine part of the overall roadwork improvements. The construction of the culvert was included in the bidding process with roadway improvements and was funded through the same Federal aid work project, so commencement or completion of the culvert was included in general news about the highway improvements. Work on road improvements and construction of Bridge No. 5722 was completed in November 1936.²⁵

²⁰ While Bridge No. 5722 was designed using the 1932 standard plan, because the bridge was skewed 35 degrees it is known as a "Special Triple 12 Ft. x 6 Ft. Box Culvert." See sheet 1, "Special Triple 12 Ft x 6 Ft Box Culvert," State of Minnesota Department of Highways, "Bridge 5722 Plans," 26 June 1936.

²¹ The intermingling of the terms "bridge" and "culvert" in the discussion can be confusing. The structure served as a "bridge" in function on TH 59 and US 63; thus, the MHD designated it as Bridge No. 5722, but it remained a culvert in engineering design and construction.

²² The contract work on US 63 and Bridge No. 5722 was one of 10 highway construction projects scheduled in nine different counties in 1936 included within Federal Aid Project 409-C. See "Let Paving Job For 4 Miles Here," *Spring Valley Tribune*, 4 June 1936.

²³ A.E. Palen, District Engineer, to N.W. Elsberg, Commissioner of Highways, 15 December 1936, Bridge No. 5722 Correspondence File, Minnesota Department of Transportation, St. Paul, Minn.

²⁴ "Weekly Bridge and Culvert Materials and Progress Report," 24 July 1936, Bridge No. 5722 Correspondence File, Minnesota Department of Transportation, St. Paul, Minn.

²⁵ E.J. Miller, photograph of completed Bridge No. 5722, 5 November 1936, in Bridge No. 5722 envelope, Box 4, Bridge Photographs and Negatives, Minnesota Highway Department Bridge Division, Minnesota State Archives, Minnesota Historical Society, St. Paul, Minn.

In 1993 the Minnesota Department of Transportation (MnDOT) proposed the replacement of Bridge No. 5722 as part of the road improvement of US 63 through Spring Valley. However, the culvert was not replaced because it was evaluated as eligible for the National Register in the “Historic Bridge Survey of Fillmore County,” a study occurring concurrently with the MnDOT project, and subsequently determined eligible by MnDOT.²⁶ Later, Bridge No. 5722 was one of 24 state-owned bridges in Minnesota identified for preservation consideration in a 2005 Programmatic Agreement.²⁷ In 2006 a Historic Bridge Management Plan for Bridge No. 5722 was prepared for MnDOT to outline the stabilization, preservation, and maintenance treatments of the culvert.²⁸

Comparative analysis

The adaptation of a standardized concrete box culvert plan as a bridge for combined vehicular and pedestrian traffic use on a U.S. Highway is unusual in Minnesota. In 2011 only two other structures on U.S. Highways in Minnesota involve a similar adaptation, primarily the use of the culvert top slab as a driving surface, neither of which resembles a bridge design and configuration like Bridge No. 5722.²⁹ Bridge No. 6568 is a four-cell culvert in Marshall, Minnesota, built in 1950, that carries a large four-lane intersection over the Redwood River, which flows diagonally underneath. Its resemblance to Bridge No. 5722 is limited to its general urban location and use of sidewalks and railings to accommodate pedestrian traffic. Structurally, it is a large, complex, custom design based generally on a box culvert configuration for this location, rather than being a direct adaptation of an existing standard plan. In that sense, it does not represent the same combination of efficiency and economy. It was designed by a consulting engineer

²⁶ Dennis Gimmestad, Government Programs and Compliance Officer, Minnesota State Historic Preservation Office, letter to Joseph Hudak, Cultural Resources Unit, Minnesota Department of Transportation, dated 21 November 1994, Bridge No. 5722 Spring Valley, Fillmore County. Available at Minnesota State Historic Preservation Office History/Architecture Inventory Files, State Historic Preservation Office, St. Paul, Minn.; Hess, Roise and Company, “Historic Bridge Survey of Fillmore County, Minnesota,” prepared for the Fillmore County Highway Department (April 1993). Available at the Minnesota State Historic Preservation Office, Minnesota Historical Society, St. Paul, Minn.

²⁷ See “Programmatic Agreement among the Federal Highway Administration, the Minnesota State Historic Preservation Office, the Advisory Council on Historic Preservation, the Department of the Army, Corps of Engineers, St. Paul District, and the Minnesota Department of Transportation regarding Implementation of the Federal-Aid Highway Program in Minnesota,” executed on June 21, 2005 (2005 Section 106 PA). See also Attachment B: “Bridges to Preserve” (including Bridge No. 5722 among the 24 enumerated bridges), in the “Section 106 Programmatic Agreement Regarding Pre-1956 Historic Bridges in Minnesota,” executed 2008 among the same agencies. Available at the Cultural Resources Unit, Minnesota Department of Transportation, St. Paul, Minn., and the Minnesota State Historic Preservation Office, Minnesota Historical Society, St. Paul, Minn.

²⁸ Historic Bridge Management Plan, Bridge No. 5722, prepared for Minnesota Department of Transportation, June 2006, available at the Cultural Resources Unit, Minnesota Department of Transportation, St. Paul, Minn.

²⁹ Data summary provided from James Pierce, MnDOT, personal communication to Mead & Hunt, Inc., 7 January 2011.

for the MHD and represents the later post-World War II period of MHD bridge specifications rather than the 1930s period of Bridge No. 5722.³⁰

Bridge No. 8654, on the other hand, was built in 1936, the same year as Bridge No. 5722, but is a diminutive structure—essentially a culvert-type structure intended to function as a pedestrian tunnel or walkway beneath US 61 in Lake City, Minnesota. The design is based on a single-cell culvert that is 5'-0" wide (as opposed to the three 12'-0" cells of Bridge No. 5722), just wide and high enough for a person to walk beneath the highway and avoid the traffic above. It is not a comparable structure to Bridge No. 5722, having a span of approximately 5'.³¹

³⁰ See Bridge No. 6568 Plan Set (10 sheets), prepared by Austin J. Kilgore, Minnesota Registered Engineer, Approved 1950 by M.O. Giertson, Bridge Engineer, and O.L. Kipp, Chief Engineer, Minnesota Department of Highways (including plan for Proposed Street Intersection, 1949), and Bridge No. 6568 Structure Inventory Report, 28 February 2011; digitized versions of plans and report available through Minnesota Department of Transportation Website: <http://dotapp3.dot.state.mn.us>, or MnDOT Bridge Office, Oakdale, Minnesota.

³¹ See Bridge No. 6568 Structure Inventory Report, 28 February 2011; digitized version available through Minnesota Department of Transportation Website: <http://dotapp3.dot.state.mn.us>, or MnDOT Bridge Office, Oakdale, Minnesota.

Part II. Structural/Design Information

A. General Statement:

1. Character: Bridge No. 5722 is a cast-in-place concrete box culvert and is significant for its unusual engineering adaptation of a standardized box culvert plan for use on a U.S. Highway to meet site-specific conditions.

2. Condition of fabric: Good.

B. Description: Bridge No. 5722 is a three-cell, concrete, cast-in-place box culvert. A culvert is traditionally defined as a drainage opening beneath a roadway with a span of less than 20 feet.³² The structural unit or hydraulic opening through which water flows is called a cell; two or more adjacent cells are combined to create larger culvert structures. The box culvert is traditionally chosen for moderate spans that carry live loads, such as vehicular traffic. Typically, culverts are unadorned, simple structures without railings or sidewalks, and in almost all cases with 2 or more feet of fill between the top of the structure and the roadway. With less than 2 feet of fill, Bridge No. 5722 is very unusual because the top of the culvert structure serves as the highway driving surface.³³ Older concrete culverts like Bridge No. 5722 were cast-in-place concrete, but since World War II these structures have been largely superseded by pre-cast concrete units, manufactured off-site. Cast-in-place concrete culverts are monolithic structures, poured into a pre-made form so the base, top, and walls of the culvert are formed as a single unit and do not have construction joints.

Bridge No. 5722 has a 1'-6"-thick concrete base, 0'-10" concrete sidewalls, and a 0'-11" top slab that also serves as the driving surface. The top slab is deepest at the driving surface and shallower beneath the sidewalks because the sidewalks do not need to support heavy vehicular loads. Each cell is 12'-0" x 6'-0" x 67'-0". The interior concrete surface of the culvert has lines and marks reflecting the original wood forms when the structure was cast. The culvert floor continues from wingwall to wingwall along a line connecting the ends of the extended cell walls, and constitutes an apron that extends 6'-8" from the deck width on both sides of the structure. The culvert walls extend the same 6'-8" distance as the culvert floor, with their height diminishing diagonally from the top of the culvert to the concrete floor. Reinforced concrete wingwalls measuring 6'-0" extend from the four corners of the culvert. The northwest wingwall is connected to a retaining wall providing support for the dead end of Courtland Street. The southwest wingwall abuts a concrete retaining wall that was added to the southwest bank of Spring Creek.

Bridge No. 5722 has an overall structure length of 49'-5" and an out-to-out deck width of approximately 67', equal to the culvert's length. The deck accommodates a 35'-4" roadway and two 6'-4" sidewalks, which are continuations of sidewalks along North Section Street. The culvert is skewed 35 degrees to accommodate the angle of the creek as it crosses beneath North Section Street.

³² U.S. Department of Transportation, *Bridge Inspectors Training Manual/90*, 19-1.

³³ Modern precast culverts, when used with minimal fill below the roadway, employ a "distribution slab" to distribute the vehicular load across the top of the culvert, a device not used in Bridge No. 5722.

Bridge No. 5722 also features ornamental metal railings along both sidewalks. The railing was specified as a "Wisconsin Railing" on the original plans. The overall railing length on each side is 47'-0" and is comprised of seven sections that are each approximately 6' long. The welded steel railing is approximately 4' high with regularly spaced square metal verticals and welded circles. Two 3'-9" concrete endposts with recessed center panels terminate the railing. No bridge plates were found on the endposts. A modern metal chain-link fence extends off the culvert from the north endpost.

C. Site Information: Bridge 5722 carries North Section Street (US 63/TH 1) over Spring Valley Creek in the city of Spring Valley, Fillmore County, Minnesota. It is located in an urban area.

Part III. Sources of Information

A. Primary Sources:

Bridge No. 5722 Correspondence File. Bridge Office. Minnesota Department of Transportation. St. Paul, Minn.

Bridge Photographs and Negatives. Minnesota Highway Department Bridge Division. Minnesota State Archives. Minnesota Historical Society, St. Paul, Minn.

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B. Secondary Sources:

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<http://onlinepubs.trb.org/onlinepubs/circulars/ec104.pdf> (accessed 16 April 2014).

Mead & Hunt, Inc., and HNTB. "Minnesota Department of Transportation Historic Bridge Management Plan, vol. 3, Bridge 5722." Prepared for Minnesota Department of Transportation (June 2006). Available at the Minnesota State Historic Preservation Office, St. Paul, Minn.

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U.S. Department of Transportation. *Bridge Inspectors Training Manual/90*. Washington D.C.: Federal Highway Administration, 1991.